Homoscedasticity and Its Testing: From Conditional Expectation to the Breusch–Pagan Test

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Part 11 / 12

The Breusch–Pagan Test: Model, Setup, and Asymptotic Properties

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1 The Breusch–Pagan Test

1.1 Motivation: what heteroscedasticity looks like

Recall the linear model

$$Y_i = \beta_0 + \beta_1 x_i + u_i, \qquad \mathbb{E}[u_i \mid x_i] = 0.$$

Under homoscedasticity, $\mathrm{Var}(u_i \mid x_i) = \sigma^2$ for all i. Under heteroscedasticity, $\mathrm{Var}(u_i \mid x_i) = \sigma^2(x_i)$ varies with x_i.

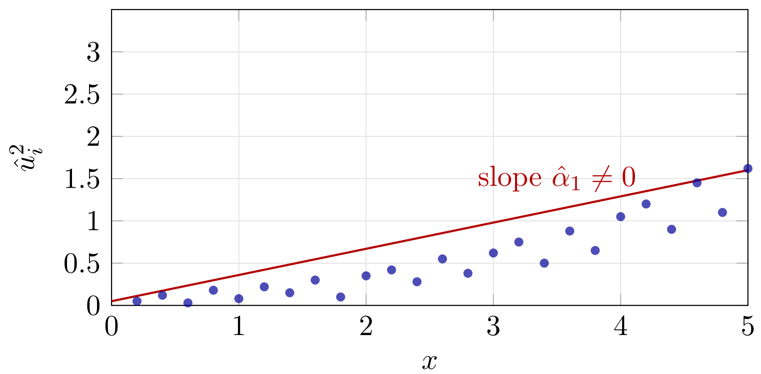
The intuition is simple: if we plot the squared residuals $\hat{u}_i^2$ against x_i and see a pattern — a trend, a funnel shape, any systematic dependence — that is evidence against homoscedasticity. The Breusch–Pagan test formalises this visual check by asking: *does the regression of $\hat{u}_i^2$ on x_i have a significantly non-zero slope?*

The difficulty is that "non-zero slope" covers infinitely many patterns. One must commit to a specific parametric family for $\sigma^2(x)$. The BP test assumes a **linear** variance function:

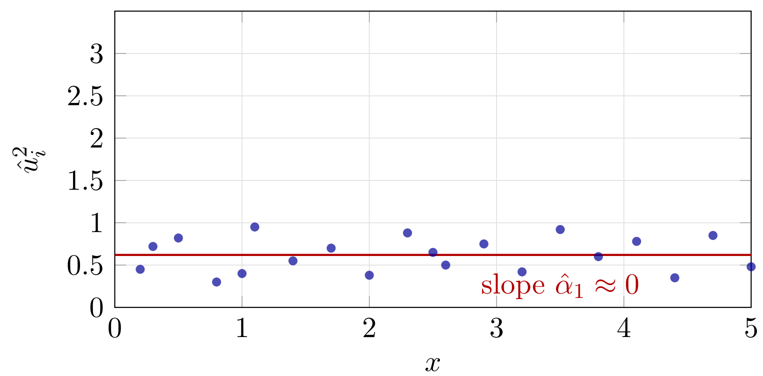
$$\mathrm{Var}(u_i \mid x_i) = \alpha_0 + \alpha_1 x_i.$$

This is simple, tractable, and sufficient to detect the most common form of heteroscedasticity (monotone dependence on x). The price is that non-linear or non-monotone patterns may be missed.

Squared residuals vs. x: heteroscedastic pattern



Squared residuals vs. x: homoscedastic pattern



1.2 Parametrization, Nuisance Parameters, and Composite Hypotheses

We follow the Probabilist's notes of June 23, 2026.

When a hypothesis does not identify the measure

A statistical model is a family of probability measures $\{P_\theta : \theta \in \Theta\}$. A hypothesis is a restriction on Θ — a subset $H \subseteq \Theta$. The hypothesis determines not a single measure but a sub-family $\{P_\theta : \theta \in H\}$.

Definition 1. A hypothesis H is **simple** if it contains a single parameter point, $H = \{\theta_0\}$, so that P_θ is uniquely identified under H. It is **composite** if $|H| > 1$.

The key situation: the parameter space factors as $\Theta = \Theta_1 \times \Theta_2$, and the hypothesis concerns only one component:

$$H_0 : \ \theta_1 = \theta_{1,0}, \qquad \text{i.e.} \quad H_0 = \{\theta_{1,0}\} \times \Theta_2 \ \subseteq \ \Theta.$$

This is a composite hypothesis whenever $|\Theta_2| > 1$. The component θ_2 is a **nuisance parameter**: it enters the model and is estimated, but is not the target of the test. It remains free under both H_0 and H_a.

Remark 2. The alternative $H_a = (\Theta_1 \setminus \{\theta_{1,0}\}) \times \Theta_2$ is also composite, for the same reason. The nuisance parameter θ_2 is unspecified under both hypotheses. This is the generic situation in practice.

Three cases

The mutual structure of H_0 and H_a in the parameter space determines the complexity of the testing problem.

Case 1: Both H_0 and H_a are simple. Both hypotheses identify a single measure. This requires the model to be fully specified under each hypothesis — no free parameters on either side. This situation is essentially theoretical; it is the setting of the Neyman–Pearson lemma, where the most powerful test exists and is given by the likelihood ratio.

Case 2: H_0 is simple, H_a is composite. The null is a single measure; the alternative is a family. This requires Θ_2 to be a singleton — the nuisance parameter is known. The structure of H_a in the parameter space matters for power, but the null distribution of any test statistic is fully determined (since there is only one null measure). This case is rare in regression: it requires knowing σ^2 (or all σ_i^2) under H_0.

Case 3: Neither H_0 nor H_a is simple. Both contain infinitely many measures. This is the generic case in regression. Now the nuisance parameter θ_2 is free on both sides, and its interaction with θ_1 — the *mutual location and structure* of H_0 and H_a in Θ — is the central difficulty. The null distribution of any test statistic depends on how θ_2 is handled (estimated, profiled out, or otherwise eliminated).

Remark 3 (the Probabilist's question). When we have parameters (θ_1, θ_2) and formulate $H_0 : \theta_1 = \theta_{1,0}$ with θ_2 unspecified, we are in Case 3: both $H_0 = \{\theta_{1,0}\} \times \Theta_2$ and $H_a = (\Theta_1 \setminus \{\theta_{1,0}\}) \times \Theta_2$ are composite. So yes: having several parameters but testing only one of them is generically Case 3.

Application to regression: three layers of parametrization

The regression model $y_i = \beta^\top x_i + u_i$ involves parameters at three distinct levels. At each level, the testing problem is Case 3, but the nuisance space shrinks as we impose more structure.

Layer 1: The regression coefficients β

The most primitive parametrization of the model treats each error variance as a free parameter:

$$\theta = (\beta,\, \sigma_1^2, \ldots, \sigma_n^2) \ \in \ \mathbb{R}^p \times \mathbb{R}^n_{>0}.$$

A hypothesis about a single regression coefficient, say $H_0 : \beta_j = 0$, leaves all of β_{-j} and all σ_i^2 free. The nuisance space is $\mathbb{R}^{p-1} \times \mathbb{R}^n_{>0}$, which has dimension $p - 1 + n$ — it grows with the sample size.

No finite-dimensional sufficient reduction exists, and no standard test statistic has a tractable null distribution without further restrictions. The t-test for $\beta_j = 0$ only becomes available after imposing homoscedasticity (Layer 2 below).

Layer 2: The error variances σ^2

Homoscedasticity replaces the n-dimensional nuisance $(\sigma_1^2, \ldots, \sigma_n^2)$ with a single scalar σ^2:

$$\theta = (\beta,\, \sigma^2) \;\in\; \mathbb{R}^p \times \mathbb{R}_{>0}.$$

The nuisance space is now finite-dimensional and does not grow with n. The OLS estimator $\hat{\beta}$ and the residual variance $\hat{\sigma}^2$ are jointly sufficient; the t- and F-statistics are pivotal under normality. This is still Case 3 (both $H_0 : \beta_j = 0$ and H_a are composite, since σ^2 is free on both sides), but the nuisance is manageable.

Testing homoscedasticity itself — $H_0 : \sigma_1^2 = \cdots = \sigma_n^2$ — cannot be formulated within this layer, because the constraint being tested is *the defining assumption of the layer*. To test it, we need the richer parametrization of Layer 1, restricted in a structured way.

Layer 3: The variance regressors α

The Breusch–Pagan test imposes a parametric form on the variances:

$$\sigma_i^2 = h(\alpha^\top z_i), \qquad \alpha = (\alpha_0, \alpha_1, \ldots, \alpha_q)^\top.$$

This replaces the n-dimensional nuisance of Layer 1 with a $(q+1)$-dimensional parameter α, making the problem tractable. The full parameter is now

$$\theta = (\beta,\, \alpha) \;\in\; \mathbb{R}^p \times \mathbb{R}^{q+1}.$$

The homoscedasticity null is

$$H_0 :\; \alpha_1 = \cdots = \alpha_q = 0,$$

which constrains q components of α to zero while leaving β and α_0 free. Under H_0, the nuisance is $(\beta, \alpha_0) \in \mathbb{R}^p \times \mathbb{R}$; under H_a, it is $(\beta, \alpha) \in \mathbb{R}^p \times \mathbb{R}^{q+1}$. Both are composite: Case 3.

The structure is now tractable, however. The score (LM) test evaluates the gradient of the log-likelihood with respect to the q constrained components $(\alpha_1, \ldots, \alpha_q)$ at the restricted MLE (= OLS under H_0). Under regularity conditions, the LM statistic has a limiting χ_q^2 distribution — with q degrees of freedom equal to the number of restrictions imposed by H_0.

Remark 4 (Degrees of freedom as a count of restrictions). The χ_q^2 in the BP test counts the number of *scalar* restrictions imposed by H_0, not the dimension of the nuisance parameter or the full parameter space. Each of $\alpha_1, \ldots, \alpha_q$ is set to zero: that is q restrictions, giving χ_q^2. In simple regression with a single variance regressor ($q = 1$), this is χ_1^2. In multiple regression with k variance regressors ($q = k$), it is χ_k^2.

The three layers are summarised in Table 1.

1.3 The Breusch–Pagan test: formal setup

Model. Linear model with error variance $\mathrm{Var}(u_i \mid x_i) = h(\alpha_0 + \alpha_1 x_i)$ for some function h. The null hypothesis is

$$H_0 : \alpha_1 = 0 \qquad \text{(variance does not depend on } x\text{)}.$$

Under H_0, $\mathrm{Var}(u_i \mid x_i) = \sigma^2 = h(\alpha_0)$: homoscedasticity.

Test procedure. Since u_i is unobservable, replace it by OLS residuals $\hat{u}_i$:

Layer	Parameter	Nuisance dimension	Testing problem
1 (β only)	$(\beta, \sigma_1^2, \ldots, \sigma_n^2)$	$p - 1 + n$ (grows)	Intractable without structure
2 (homosc.)	(β, σ^2)	p (fixed)	t/F tests; pivot exact
3 (BP)	$(\beta, \alpha_0, \alpha_1, \ldots, \alpha_q)$	$p + 1$ (fixed)	LM test; χ_q^2 asymp.

Table 1: Three layers of parametrization in the BP regression model. At each layer the testing problem is Case 3 (both H_0 and H_a composite), but the nuisance space shrinks from $O(n)$ to a fixed finite dimension, making standard inference feasible.

1. Fit OLS to obtain residuals $\hat{u}_i$.
2. Run the auxiliary OLS regression of $\hat{u}_i^2$ on x_i (and a constant).
3. Compute nR^2 from this auxiliary regression.
4. Under H_0: $nR^2 \xrightarrow{d} \chi_1^2$ (χ_k^2 for k regressors).
5. Reject H_0 at level α if $nR^2 > \chi^2_{1,\alpha}$.

The statistic nR^2 is the LM statistic evaluated under H_0: its form as n times the R^2 of the auxiliary regression is a consequence of applying the general LM principle to the normal likelihood, where the score and Fisher information take a particularly simple form.

Remark 5. The χ^2 distribution is asymptotic. The original Breusch–Pagan test assumed normality of u; the version due to Koenker (the "studentised" BP test) does not require normality and is more robust in small samples.

1.4 Multivariate model setup and the nature of H_0

We follow the Probabilist's notes of June 21, 2026.

The regression model

The general linear regression model with n observations and p regressors is

$$\begin{cases} y_1 = \beta^\top x_1 + u_1 \\ \quad\vdots \\ y_n = \beta^\top x_n + u_n, \end{cases}$$

where each regressor vector is a column:

$$\begin{cases} x_1 = (x_{11}, \ldots, x_{1p})^\top \\ \quad\vdots \\ x_n = (x_{n1}, \ldots, x_{np})^\top. \end{cases}$$

The errors are independently normally distributed:

$$u_i \;\sim\; \text{indep}\, \mathcal{N}(0,\, \sigma_i^2),$$

so that in vector form

$$u = (u_1, \ldots, u_n)^\top \;\sim\; \mathcal{N}\big(0_n,\; \text{diag}(\sigma_1^2, \ldots, \sigma_n^2)\big).$$

Parametric variance structure

The main modelling assumption of the BP test: each error variance can be written as

$$\sigma_i^2 \;=\; h(\alpha^\top z_i),$$

where $h(\cdot)$ is a twice-differentiable positive function, $\alpha = (\alpha_0, \alpha_1, \ldots, \alpha_q)^\top$ is the *variance parameter vector*, and $z_i = (1, z_{i1}, \ldots, z_{iq})^\top$ is an observable vector of *variance regressors* (often $z_i = x_i$, but not necessarily).

The null hypothesis and its character

The homoscedasticity null is

$$H_0 : \;\alpha_1 = \alpha_2 = \cdots = \alpha_q = 0,$$

i.e. all slope components of α are zero. Under H_0, $\sigma_i^2 = h(\alpha_0) =: \sigma^2$ for all i — a constant.

Remark 6 (The null is not a simple hypothesis). The Probabilist observes: H_0 is *not* of the form "the parameter equals a specific value." It is a *structural restriction* on the parameter vector: it sets q components of α to zero while leaving α_0 (and all of β) unrestricted. The restricted parameter space is therefore a proper subspace of dimension strictly less than the full parameter space — but it is not a single point. Hence H_0 is a **composite** hypothesis, not a simple one. This distinction matters: the LM (score) test statistic has a χ_q^2 limiting distribution under H_0 precisely because q restrictions are imposed, and the χ^2 degrees of freedom count the number of restrictions, not the dimension of the parameter space.

1.5 Conditional distribution of u_i and the status of σ_i^2

We follow the Probabilist's notes of June 24, 2026.

Notation: intercept convention and indexation

We clarify the indexation of regressors. The coefficient vector β and the regressor vector x_i each have $p + 1$ components indexed from 0:

$$\beta = (\beta_0, \beta_1, \ldots, \beta_p)^\top, \qquad x_i = (x_{i0}, x_{i1}, \ldots, x_{ip})^\top, \qquad x_{i0} \equiv 1 \text{ (intercept)}.$$

Then $\beta^\top x_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$. Similarly, the variance regressors satisfy

$$z_i = (z_{i0}, z_{i1}, \ldots, z_{iq})^\top, \qquad z_{i0} \equiv 1,$$

so $\alpha^\top z_i = \alpha_0 + \alpha_1 z_{i1} + \cdots + \alpha_q z_{iq}$. The intercept component in each vector absorbs the constant term and is *not* a free parameter to be tested.

Remark 7 (Consistency with earlier notation). In the Multivariate model setup subsection (§ 1.4) we wrote $x_i \in \mathbb{R}^p$. With the intercept convention, the dimension is $p+1$. No conflict arises if we interpret p there as the total number of columns including the intercept column. The present subsection adopts the explicit 0-indexed convention throughout.

Random variances: the problem

The Probabilist raised the following issue. In the BP model, $\sigma_i^2 = h(\alpha^\top z_i)$. If z_i is itself a random vector (e.g. the observed regressors on the i-th unit), then σ_i^2 is a *random variable*, not a fixed parameter. Random variances are not parameters in the classical sense, and the log-likelihood cannot be written in the usual form.

This is a genuine tension, and it is resolved — not avoided — by working with *conditional* distributions.

Conditional probability as a random variable

Fix observation i. The conditional probability of the event $\{u_i \in A\}$ given Z_i is defined as

$$P\{u_i \in A \mid Z_i\} \;=\; \mathbb{E}\big[\mathbf{1}_{u_i \in A} \mid Z_i\big],$$

which is a $[0,1]$-valued random variable — measurable with respect to $\sigma(Z_i)$ — defined up to a P-null set. Here Z_i stands for either the random vector itself or its generated σ-algebra; the two phrasings are equivalent.

Remark 8. This is conditioning with respect to a *random variable*, not with respect to an event. In particular, $P\{u_i \in A \mid Z_i = z\}$ for a fixed value $z \in \mathbb{R}^{q+1}$ requires care: the event $\{Z_i = z\}$ has probability zero when Z_i is continuous, so the conditional probability cannot be defined as a ratio. The rigorous object is the *regular conditional distribution*, defined below.

The kernel representation

For P-almost every ω, the map $A \mapsto P\{u_i \in A \mid Z_i\}(\omega)$ is a probability measure on $\mathbb{R}$. Writing $z = Z_i(\omega)$, we define the **conditional distribution kernel**

$$\mu_i(z,\, A) \;:=\; P\{u_i \in A \mid Z_i\}(\omega),$$

so that $\mu_i\colon \mathbb{R}^{q+1} \times \mathcal{B}(\mathbb{R}) \to [0,1]$. For every fixed $z \in \mathbb{R}^{q+1}$, the map $A \mapsto \mu_i(z, A)$ is a probability measure on $\mathbb{R}$ (the distribution of u_i given $Z_i = z$). For every fixed Borel set A, the map $z \mapsto \mu_i(z, A)$ is measurable.

Remark 9 (Deterministic map from $\mathbb{R}^{q+1}$ to measures). The Probabilist's key observation: what we actually need is not the full random-measure object $\omega \mapsto \mu_i(Z_i(\omega), \cdot)$, but the *deterministic* function $z \mapsto \mu_i(z, \cdot)$. This function maps each fixed vector $z \in \mathbb{R}^{q+1}$ to a specific probability measure on $\mathbb{R}$. The randomness of Z_i only matters when we integrate over it — which we do not need for the likelihood computation, where z_i are treated as fixed design points.

The BP distributional assumption

Under the simplification of Breusch and Pagan, the conditional distribution of each u_i given $Z_i = z$ is normal with mean zero and variance $h(\alpha^\top z)$:

$$\mu_i(z, \cdot\,) \;=\; \mathcal{N}\big(0,\; h(\alpha^\top z)\big).$$

In the notation of the random-variable form:

$$u_i \mid Z_i \;\sim\; \mathcal{N}\big(0,\; h(\alpha^\top Z_i)\big).$$

The full statement in the original Breusch–Pagan paper is stronger: the entire error vector u has a *joint* Gaussian conditional distribution given $Z = (z_1, \ldots, z_n)$:

$$u \mid Z \;\sim\; \mathcal{N}\Big(0_n,\; \operatorname{diag}\big(h(\alpha^\top z_i)\big)_{i=1}^n\Big).$$

The diagonal covariance encodes both normality of each u_i and mutual independence of $u_1, \ldots, u_n$ given Z.

Remark 10 (From conditional to "unconditional"). The Probabilist notes that writing the distribution of u without conditioning on Z is informal: when Z_i is a continuous random vector, $P(Z_i = z_i) = 0$ for any fixed $z_i \in \mathbb{R}^{q+1}$, so the conditional statement $u \mid Z_i = z_i$ cannot be interpreted as ordinary conditional probability with respect to an event. The rigorous sense is exactly the kernel $\mu_i(z_i, \cdot)$ constructed above. For the likelihood this is harmless: the log-likelihood is written as a function of the observed z_i values, treating them as deterministic inputs, which corresponds to the standard fixed-regressor convention in regression analysis.

Remark 11 (Do we need random measures?). The Probabilist's notes acknowledge a mild self-contradiction: the stated goal was to avoid random measures, yet the conditional probability $\omega \mapsto \mu_i(Z_i(\omega), \cdot)$ is precisely a random measure. The resolution is pragmatic: we invoke random measures only to *justify* the existence and measurability of the kernel $\mu_i(z, \cdot)$ (this is the content of the regular conditional distribution theorem), after which we work entirely with the deterministic function $z \mapsto \mu_i(z, \cdot)$ and need not refer to random measures again.

The incidental parameters problem

We follow the Probabilist's notes of June 25, 2026.

Returning to the free-variance model $u_i \sim \text{indep}\,\mathcal{N}(0, \sigma_i^2)$: the Probabilist notes that in this formulation there is *no way to estimate the σ_i^2 consistently.* Each observation u_i (equivalently, each residual $\hat{u}_i$) is a single data point drawn from $\mathcal{N}(0, \sigma_i^2)$; a new observation u_{n+1} arrives with its own fresh σ_{n+1}^2 and tells us nothing about the previous variances. The number of free parameters grows with n, so the MLE of σ_i^2 cannot converge.

Remark 12 (Incidental parameters). This is the classical *incidental parameters problem* (Neyman and Scott, 1948). The σ_i^2 are "incidental" parameters — one per observation — as opposed to "structural" parameters such as β, which are shared across all observations. The MLE of incidental parameters is generally inconsistent and, crucially, its inconsistency contaminates the MLE of the structural parameters too. The BP parametrisation $\sigma_i^2 = h(\alpha^\top z_i)$ is precisely the fix: it replaces n incidental parameters with $q + 1$ structural ones, restoring consistency.

The Probabilist also notes the connection to Bayesian hierarchical models: imposing $\sigma_i^2 = h(\alpha^\top z_i)$ with z_i i.i.d. is analogous to placing a parametric prior on the σ_i^2 and treating the shared parameter α as a hyperparameter.

Does $z_i = x_i$ imply homoscedasticity?

Boris raised the following question. The authors of the BP paper remark that often $z_i = x_i$ is a natural choice. If in addition x_i are i.i.d., then $\sigma_i^2 = h(\alpha^\top x_i)$ are i.i.d. random variables. Does this make the u_i i.i.d., and therefore homoscedastic?

The Probabilist's answer: no. Homoscedasticity as defined in our earlier chapters requires the conditional variance to be both *constant* and *deterministic*:

$$\operatorname{Var}(u_i \mid x_i) \;=\; \sigma^2 \quad \text{(the same fixed real number for all } i\text{).}$$

Even if the random variables $\sigma_i^2 = h(\alpha^\top x_i)$ are identically distributed — which they are when x_i are i.i.d. — they are still random, not constant. Identity in distribution is strictly weaker than equality as numbers.

Remark 13 (Gradations of "equal variance"). The Probabilist identifies two intermediate statements between "i.i.d." and homoscedasticity:

1. *Identically distributed:* $\sigma_1^2 \stackrel{d}{=} \sigma_2^2 \stackrel{d}{=} \cdots$ (same law, but they may take different values on any given ω). Not homoscedasticity.

2. *Pointwise equal:* $\sigma_1^2(\omega) = \sigma_2^2(\omega)$ for all $\omega \in \Omega$. This is a much stronger statement — it means the random variables are modifications of each other — but it is still not homoscedasticity, because the common value $\sigma^2(\omega)$ is still random and not a constant.

The Probabilist notes that statement (2) is hard to imagine in any realistic model and is cited here only for conceptual completeness. The upshot: randomness of σ_i is not an obstacle to be eliminated but a feature that *defines* heteroscedasticity in the random-regressor framework.

Conditioning on z versus conditioning on x

The Probabilist raises a further subtlety. Our definition of homoscedasticity throughout these notes conditions on the regression regressors x_i:

$$\mathrm{Var}(u_i \mid x_i) = \sigma^2.$$

The BP test, however, parametrises variance through the auxiliary vector z_i, which may differ from x_i. The condition being tested is therefore

$$\mathrm{Var}(u_i \mid z_i) = \sigma^2,$$

a statement about a different conditioning sigma-algebra. When $z_i \neq x_i$ (e.g. z_i contains functions of x_i or additional covariates), the BP test does not directly test our earlier homoscedasticity definition.

The BP paper does not address this gap. As the Probabilist puts it: it is swept under the rug. A careful treatment would require either (a) taking $z_i = x_i$ explicitly, in which case both definitions coincide; or (b) reformulating homoscedasticity relative to the larger σ-algebra $\sigma(x_i, z_i)$. We flag this as an open point in our exposition.

Remark 14 (Reference gap). The Probabilist consulted the *International Encyclopedia of Statistics* (multi-author, ∼2016, over 1700 pages). The entry on heteroscedasticity did not provide a formal definition of homoscedasticity. This is symptomatic of a broader pattern in the statistics literature: the term is used freely without a measure-theoretic anchor.

A note on the institutional gap between probability and statistics

Boris observes: having studied statistics at three Canadian universities (SFU, Western, UofT), he found that only UofT had probability-specialist faculty and included measure-theoretic probability as a standard component of the PhD curriculum. At most departments the typical PhD statistician completes their degree without ever having taken a graduate-level probability course. From this Boris draws the inference: expecting rigorously probability-grounded arguments in mainstream statistics papers is structurally unrealistic, because the institutional pipeline does not produce researchers with those tools.

The observation extends to the disciplinary split Boris notes: probability theory has penetrated quantitative finance far more deeply than it has statistics itself. Stochastic calculus, risk-neutral pricing, and measure-theoretic foundations of martingale theory became standard in financial mathematics from the 1970s onward, driven by Black–Scholes and the Itô calculus. In contrast, most working statisticians operate with a toolkit grounded in Fisher, Neyman, and Pearson — a framework that predates Kolmogorov's axiomatisation and was never reformulated in its terms.

Remark 15 (Claude's observation). Boris's question — whether probability is penetrating more areas or retreating — does not have a single answer. The picture bifurcates. On one side: machine learning and modern data science have *reduced* the role of measure-theoretic probability, replacing it with finite-sample concentration inequalities and computational heuristics. A practitioner training neural networks operates entirely outside Kolmogorov's framework. On the other side: areas such as nonparametric Bayes (Gaussian processes, Dirichlet processes), causal inference under interference, and robust statistics have *increased* their reliance on advanced probability.

The finance example is worth unpacking. Measure theory entered finance not because financial economists sought mathematical rigour for its own sake, but because the problems demanded it: pricing derivatives required integrating over continuous paths, and the only tools for that were stochastic integration and change-of-measure arguments. The tool followed the problem. Statistics, by contrast, has problems (conditional independence, exchangeability, nonparametric priors) that equally demand measure theory — but the discipline's culture and training infrastructure have not converged to the same conclusion.

Whether this changes depends less on abstract intellectual trends than on institutional incentives: hiring committees, qualifying exam syllabi, and journal referee expectations. Boris's experience across three universities is a precise diagnostic of where those incentives currently sit.

1.6 The Koenker (1981) modification: non-conditional parametrization

We follow Boris's notes of June 27, 2026 (note 52).

A paper the Probabilist brought

The Probabilist brought the paper: Roger Koenker, "A note on studentizing a test for heteroscedasticity," *Journal of Econometrics* 17 (1981), 107–112. The paper modifies the assumptions of the BP test and proposes a small change to the test statistic.

The product parametrization

Koenker works with the following model, which the Probabilist notes avoids conditional distributions entirely:

$$y_i = x_i\beta + u_i, \tag{1.1}$$

$$u_i = \sigma_i\varepsilon_i, \tag{1.2}$$

$$\sigma_i = 1 + h(z_i\gamma), \tag{1.3}$$

where x_i and z_i are row vectors of known constants (fixed regressors), h is a smooth positive function with $h(0) = 0$, and ε_i are i.i.d. with mean 0, variance σ^2, and finite fourth moment. The null hypothesis is $H_0 : \gamma = 0$, under which $\sigma_i = 1$ and $u_i = \sigma\varepsilon_i$ is homoscedastic.

Remark 16 (Product model vs. conditional normal). BP's original model specifies a conditional normal distribution for u_i given z_i: $u_i \mid z_i \sim \mathcal{N}(0, h(\alpha^\top z_i))$. Koenker's product model (1.2) makes no normality assumption on ε_i; the distribution is an unspecified F with finite fourth moment. The conditional distribution of u_i given z_i in Koenker's model is a scaled copy of F, not necessarily Gaussian. Note also that (1.3) parametrises the *standard deviation* σ_i, not the variance: $\sigma_i^2 = (1 + h(z_i\gamma))^2 \approx 1 + 2h(z_i\gamma)$ near $\gamma = 0$.

The test statistic and the kurtosis problem

BP base their test on the OLS residuals $\hat{u} = [I - X(X'X)^{-1}X']u$, the variance estimate $\hat{\sigma}^2 = \hat{u}'\hat{u}/n$, and the centred squared residuals $\hat{w}_i = \hat{u}_i^2 - \hat{\sigma}^2$. The BP test statistic is

$$\hat{\xi} \;=\; \tfrac{1}{2}\hat{w}'Z(Z'Z)^{-1}Z'\hat{w} \,/\, \hat{\sigma}^4.$$

Koenker's theorem shows that, without the Gaussian assumption on ε, the rescaled statistic $\xi^* = \hat{\xi}/\lambda$ converges in distribution to a non-central $\chi^2_p(\eta)$, where $\lambda = \phi/(2\sigma^4)$ and $\phi = \mathrm{Var}(\varepsilon^2)$. Under Gaussian ε, $\phi = 2\sigma^4$ so $\lambda = 1$ and the size is correct. Under non-Gaussian ε, $\lambda \neq 1$ and the BP test has **asymptotically incorrect size**: the nominal α level is not achieved.

The fix Koenker proposes: replace $2\hat{\sigma}^4$ in the denominator by $\hat{\phi} = (1/n)\sum_i \hat{w}_i^2$. Since $\hat{\phi} \to \phi$ in probability, the modified statistic $\hat{\alpha}'Z'Z\hat{\alpha}/\hat{\phi}$ has the correct limiting χ^2_p distribution under H_0 without Gaussianity. Koenker notes that this modified statistic equals nR^2, where R^2 is the *uncentred* coefficient of determination in the auxiliary regression of $\hat{w}$ on Z.

Boris's observation: where are the type I and II error statements?

Boris notes a gap that applies to both the original BP paper and Koenker's note: neither paper contains an explicit statement about type I or type II error probabilities of the test. What both papers provide is an asymptotic distribution result for the test *statistic* under H_0 (and, in Koenker, under contiguous alternatives).

The Probabilist confirms: the asymptotic distribution of the test statistic is what is proved. Mention of power appears near the end of Koenker's paper, but without exact statements or computations.

Remark 17 (Distribution of the statistic is not the same as error probability control). Knowing that $\hat{\xi} \xrightarrow{d} \chi_p^2$ under H_0 is a necessary condition for the test to have asymptotically correct size, but it is not sufficient on its own. For the rejection rule "reject if $\hat{\xi} > \chi_p^2(1-\alpha)$" to have limiting type I error α, one additionally needs the convergence to be sufficiently uniform over the null (or at least to hold for all $\theta \in H_0$) and needs the quantile of the limit distribution to be a continuity point. These are standard but non-trivial conditions that the econometrics literature commonly treats as understood. Boris's observation is that they are, strictly speaking, not stated. The gap the Probabilist and Boris identify here connects directly to the foundational questions in note 53.

1.7 Asymptotic test behaviour and the notion of consistency

We follow Boris's notes of June 27, 2026 (note 53).

What "the test" actually is

When we casually say "a test," we sweep under the rug that in fact it is a *parametrized family* of tests. For the tests we encounter in practice, the family is parametrized by at least two quantities: the significance level α and the sample size n. A test at level α for sample size n is a measurable function $\phi_{n,\alpha}\colon \mathcal{X}^n \to \{0,1\}$ (reject or not), so what we call "the BP test" is really the collection $\{\phi_{n,\alpha} : n \geq 1,\, \alpha \in (0,1)\}$.

Type I and II error probabilities

For a fixed test $\phi_{n,\alpha}$ and a fixed parameter value θ:

$$\begin{aligned} \alpha_n(\theta) &:= P_\theta(\phi_{n,\alpha} = 1), \quad \theta \in H_0, \\ \beta_n(\theta) &:= P_\theta(\phi_{n,\alpha} = 0), \quad \theta \in H_a. \end{aligned}$$

When both hypotheses are simple (single measures), α_n and β_n are single numbers. When H_0 or H_a is composite, $\alpha_n(\theta)$ and $\beta_n(\theta)$ depend on the specific θ, and there is no single "the α" or "the β": they are functions on the respective hypothesis sets.

In practice one controls the *supremum*:

$$\sup_{\theta \in H_0} \alpha_n(\theta) \;\leq\; \tilde{\alpha}_n \qquad \text{and} \qquad \sup_{\theta \in H_a} \beta_n(\theta) \;\leq\; \tilde{\beta}_n.$$

These upper bounds $\tilde{\alpha}_n$ and $\tilde{\beta}_n$ are the quantities one can actually quote as guarantees.

Consistency of a sequence of tests

The asymptotic analog of the above: choose a sequence of tests ϕ_{n,α_n} with the goal that both error bounds converge to zero:

$$(\tilde{\alpha}_n,\, \tilde{\beta}_n) \;\to\; (0,0) \quad \text{as } n \to \infty.$$

A sequence of tests achieving this is called **consistent**. The Probabilist notes that this is how consistency of a test should be understood in the composite-hypothesis setting: not convergence of a single number, but convergence of a pair of uniform bounds.

Remark 18 (Why α_n may need to depend on n). The Probabilist cites a source in which Fisher argued it is unreasonable to fix the same α level for all sample sizes. The intuition: at very large n, one can afford a smaller α (the test has more power for any fixed alternative), so insisting on $\alpha = 0.05$ for $n = 10^6$ may be wasteful or even misleading. Allowing $\alpha_n \to 0$ as $n \to \infty$ — while ensuring $\beta_n \to 0$ as well — is a more honest calibration. The pair (α_n, n) replaces the single parameter α as the fundamental specification.

The asymmetry of H_0 and H_a

The Probabilist and Boris discuss why the standard framework treats α and β asymmetrically. The standard undergraduate procedure is: (1) choose the α-level parameter of the test (implemented through a threshold on the statistic); (2) β then follows from the sample size and the specific alternative — it is not chosen. In general, for a compound H_a, there is no single β to choose: it is a function of $\theta \in H_a$.

Boris traces this asymmetry to Fisher. The conceptual origin: H_0 is the *status quo* — a position held for years — while H_a is a *challenger*. We give the challenger a lower chance of survival by requiring strong evidence to reject H_0. This framing predates Kolmogorov's 1933 axiomatisation; Fisher lived until 1962 and could in principle have encountered measure-theoretic probability, but his framework was never reformulated in those terms. The Neyman–Pearson elaboration formalised β as a second parameter, but the underlying asymmetry persists.

Remark 19 (Structural reason H_0 is more often simple). Boris observes: historically H_0 is more often a simple hypothesis than H_a. This is not arbitrary. A natural H_0 is often a specific structural claim — independence, equality of means, zero coefficient — which pins down the parameter exactly (up to nuisance, which is then either known or eliminated by a pivot). The alternative, being the negation, is typically composite: "not independent," "unequal means," "non-zero coefficient." Because H_0 is frequently simple, the literature on α-level control is richer and more explicit, while power analysis (β) requires specifying a point in H_a and is treated as a secondary, design-stage consideration.

1.8 BP as a semi-parametric test: the difficulty of error bounds

We follow Boris's notes of June 29, 2026 (note 54).

Classification vs. hypothesis testing: why consistency is harder here

The Probabilist notes that α and β recall binary classification problems. The analogy is instructive precisely because it breaks down at a key point.

In a binary classification problem one typically has an *overall* distribution on the sample space: $P = \pi P_0 + (1 - \pi)P_1$, where π is the base rate (prior probability of class 1). Given this, one can define an overall misclassification rate — a single number combining the frequency of 0s and 1s — and minimise it. Consistency of a classifier can be formulated as convergence of that rate to its Bayes minimum.

In hypothesis testing there is no such overall measure. The sample comes from *either* the H_0-family of measures *or* the H_a-family — not from a mixture. No π is given, and in the frequentist framework none is sought. Type I and type II errors are therefore irreducibly separate quantities, each defined with respect to a different family of measures. This is why consistency of a test sequence cannot be defined as convergence of a single number; the $(\tilde{\alpha}_n, \tilde{\beta}_n) \to (0, 0)$ formulation from § 1.7 is the correct two-dimensional substitute.

Why worst-case error bounds are hard for the BP test

The Probabilist characterises the BP test as *semi-parametric*: the model specifies a parametric form for the conditional distribution of Y given (X, Z), but leaves the marginal distribution of (X, Z) entirely unspecified. In the fixed-regressor convention of Koenker's paper, x_i and z_i are treated as known constants, so the question of the distribution of X does not arise. But when X and Z are genuinely random, their joint distribution becomes an infinite-dimensional nuisance parameter.

Remark 20 (Why this matters for α and β). In the fixed-regressor framework, the type I error of the BP test is controlled asymptotically by the χ^2_q limit: the design conditions (A.1) of Koenker's paper ensure this uniformly over the finite-dimensional nuisance (β, α_0). The sup over all possible fixed design sequences satisfying (A.1) is bounded.

In the random-regressor framework, the sup over all joint distributions of (X, Z) is a much larger object. Without restricting that distribution (e.g. to satisfy moment conditions or tail bounds), one cannot bound $\tilde{\alpha}_n$ uniformly. This is the semi-parametric difficulty Boris identifies: the worst-case α and β are not well-defined without specifying a class of distributions for X. The BP paper and Koenker's note sidestep this by treating regressors as fixed — a convention that handles the problem by assumption rather than proof.

1.9 Return to the model: test statistic construction and the p-value gap

We follow Boris's notes of June 29 – July 1, 2026 (note 55).

The model in Koenker's formulation

We restate the model cleanly. There exist deterministic unknown parameter vectors

$$\beta = (\beta_0, \beta_1, \dots, \beta_p)^\top, \qquad \alpha = (\alpha_0, \alpha_1, \dots, \alpha_q)^\top,$$

and random variables (Y_i, X_i, Z_i), all observed, related by

$$\begin{aligned} &Y_i = \beta^\top X_i + U_i, \\ &U_i = \sigma_i \, \varepsilon_i, \qquad \varepsilon_i \text{ i.i.d.}, \quad \mathbb{E}[\varepsilon_i] = 0, \quad \mathrm{Var}(\varepsilon_i) = \sigma^2, \\ &\sigma_i^2 = h(\alpha^\top Z_i). \end{aligned}$$

The random variables U_i, σ_i, and ε_i are *not* observed. The parameters α and β are deterministic and unknown. No relationship between X_i and Z_i is assumed; they may coincide or be entirely separate.

Homoscedasticity and the null hypothesis

Homoscedasticity requires two things of σ_i:

1. **Non-randomness:** σ_i is a constant, not a random variable (it does not vary with the draw ω).
2. **Constancy:** σ_i is the same value for all i (follows from non-randomness if the parametric form $h(\alpha^\top Z_i)$ is used, since $h(\alpha^\top z)$ is deterministic when z is a fixed vector and α is deterministic).

In terms of the parameter α, these conditions are equivalent to

$$\alpha \;=\; (\alpha_0, 0, \dots, 0)^\top,$$

i.e. the null hypothesis is

$$H_0 : \; \alpha_1 = \alpha_2 = \cdots = \alpha_q = 0.$$

Note that β is unconstrained under H_0: we test the variance structure, not the mean. Also note that α_0 is free under H_0 — it determines the common variance $\sigma^2 = h(\alpha_0)$.

Remark 21 (Equivalence caveat). Boris notes the equivalence between "σ_i constant" and "$\alpha_1 = \cdots = \alpha_q = 0$" is not entirely obvious for general h and general Z_i. For example, if Z_i always lies in a subspace where $h(\alpha^\top z) = h(\alpha_0)$, the slope components of α are irrelevant even when non-zero. For the generic case — Z_i spanning $\mathbb{R}^{q+1}$ in the sense of condition (A.1) of Koenker — the equivalence holds.

Construction of the test statistic

The test proceeds in two stages.

Stage 1: OLS residuals. Estimate β by OLS on the (Y_i, X_i) pairs:

$$\hat\beta = (X^\top X)^{-1} X^\top Y, \qquad \hat U_i = Y_i - X_i^\top \hat\beta.$$

Note that $\hat\beta$ itself is discarded after this step; only the residuals $\hat U_i$ are used further.

Stage 2: Auxiliary regression. Compute the pooled variance estimate

$$\hat\sigma^2 \;=\; \frac{1}{n}\sum_{i=1}^{n} \hat U_i^2,$$

and the normalised squared residuals

$$g_i \;=\; \frac{\hat U_i^2}{\hat\sigma^2}.$$

The BP paper uses the notation g_i (the Probabilist notes it is not an especially descriptive name). Observe that $\frac{1}{n}\sum_i g_i = 1$ by construction, so the g_i are centred at 1. Under H_0 and for large n, $g_i \approx \varepsilon_i^2/\sigma^2$, which has mean 1 and is uncorrelated with Z_i.

Run the auxiliary OLS regression of g_i onto Z_i:

$$g_i = Z_i^\top \hat\gamma + \text{residual}.$$

The test statistic is *half the explained sum of squares* of this regression:

$$T_n \;=\; \tfrac{1}{2}\,\widehat{\mathrm{ESS}}, \qquad \widehat{\mathrm{ESS}} = \hat\gamma^\top (Z^\top Z)\,\hat\gamma.$$

Under H_0 and the Gaussian assumption, $T_n \xrightarrow{d} \chi_q^2$.

Remark 22 (Degrees of freedom: χ_q^2 not χ_{q-1}^2). Boris's note states the limit is χ_{q-1}^2. This appears to be a slip in counting. The null imposes q constraints ($\alpha_1 = \cdots = \alpha_q = 0$), so the LM statistic has q degrees of freedom. The vector Z_i has $q+1$ components (including the intercept $z_{i0} \equiv 1$), but the intercept column in the auxiliary regression absorbs α_0, which is *not* constrained — leaving exactly q free directions being tested. Koenker's theorem confirms: the limit distribution is χ_p^2 in his notation, where p is the dimension of γ (the slope part of α) — i.e. q in ours.

From the distribution of T_n to a p-value: the remaining gap

Boris adds (July 1, 2026): even for α-level purposes, proving $T_n \xrightarrow{d} \chi_q^2$ under H_0 is not sufficient on its own. Two additional steps are required.

Step A: Connecting the statistic to α. The rejection rule is "reject H_0 if $T_n > c_{q,1-\alpha}$," where $c_{q,1-\alpha}$ is the $(1-\alpha)$-quantile of χ_q^2. The convergence $T_n \xrightarrow{d} \chi_q^2$ implies

$$P_{H_0}(T_n > c_{q,1-\alpha}) \;\to\; \alpha \quad \text{as } n \to \infty,$$

provided $c_{q,1-\alpha}$ is a continuity point of the χ_q^2 distribution (it is) and the convergence holds uniformly enough over H_0. This step is implicit in both the BP paper and Koenker's note — it is treated as "obvious from the convergence in distribution." Boris's point is that it is not trivial and is not explicitly stated.

Step B: The p-value. Boris notes that the p-value $p_n = P(\chi^2_q \geq T_n^{\text{obs}})$ is in principle well-defined here because the limiting distribution is free of all nuisance parameters (β and α_0 drop out of the limit). This is a non-trivial property — it is the Wilks phenomenon for LM tests — and it holds because the score evaluated at the restricted MLE is asymptotically normal with covariance that does not depend on the nuisance.

However, Boris's broader point stands: for more complicated hypothesis structures (non-standard boundary cases, non-smooth parametrizations, or non-interior null points), the p-value may fail to exist in the usual sense, since there is no single scalar sufficient to rank evidence against H_0. The BP test happens to be in the favourable regime — one-sided in each component of α aggregated into a single LM quadratic form — but this should be stated, not assumed.

1.10 Why the LM statistic is χ^2: centration and nuisance

We follow Boris's notes of July 2, 2026 (note 56).

Boris's question

The standard argument that a score statistic is asymptotically χ^2_q under H_0 runs as follows.

1. **Centration.** The score identity states that under the true measure P_θ,

$$E_\theta\left[\frac{\partial}{\partial\theta}\log p(X;\theta)\right] = 0.$$

   So the score is centered at zero under the native measure.

2. **CLT.** With i.i.d. observations, the sum of individual scores obeys the CLT, and

$$\frac{1}{\sqrt{n}}\sum_{t=1}^{n}\frac{\partial}{\partial\theta}\log p(x_t;\theta) \xrightarrow{d} \mathcal{N}(0,\, \mathcal{J}(\theta)).$$

3. **χ^2.** A centered normal divided by its standard deviation, squared, is χ^2_1; the q-dimensional version gives χ^2_q.

Boris's worry: in the BP setting the measure is not fully specified under H_0 — the nuisance parameters (β, α_0) are free. So "the native measure" does not exist as a single P_{θ_0}. The score evaluated at the restricted MLE $\hat\theta = (\hat\beta, \hat\alpha_0, 0)$ uses estimated nuisance parameters, not the true ones. Does centration survive? And if the score is not centered, the CLT produces a non-zero mean, the sum of squares drifts positive, and the test over-rejects — no χ^2.

Answer: centration holds pointwise, substitution error is negligible

Step 1: The score identity is pointwise in θ. For each fixed $\theta_0 = (\beta_0, \alpha_{0,0}, 0) \in H_0$,

$$E_{\theta_0}\left[\frac{\partial}{\partial\alpha_j}\log p(x_t;\theta_0)\right] = 0, \qquad j = 1, \dots, q.$$

This holds by the usual differentiation-under-the-integral argument and requires no knowledge of the nuisance values — only that θ_0 is the true parameter. Centration is a property of the score at the *true* θ, not at an estimated one.

The difficulty Boris identifies is genuine: we cannot evaluate $d_{\alpha_j}(\theta_0)$ because θ_0 is unknown. Instead we substitute $\hat{\theta}$ (the constrained MLE under H_0), obtaining $\hat{d}_{\alpha_j} = d_{\alpha_j}(\hat{\theta})$. Decompose:

$$\hat{d}_{\alpha_j} \;=\; \underbrace{d_{\alpha_j}(\theta_0)}_{\text{centered, mean 0}} \;+\; \underbrace{\left[d_{\alpha_j}(\hat{\theta}) - d_{\alpha_j}(\theta_0)\right]}_{\text{estimation error}}.$$

The first term is centered. The second is the price of using estimated nuisance.

Step 2: Block-diagonal structure kills the estimation error. The Fisher information matrix $\mathcal{J}$ is block-diagonal between (β, α_0) and $(\alpha_1, \ldots, \alpha_q)$:

$$\mathcal{J}_{\alpha_j,\beta} = 0, \qquad \mathcal{J}_{\alpha_j,\alpha_0} = 0,$$

as computed in our earlier example (the cross-derivative contains $E[u_t] = 0$). This means the constrained MLE satisfies

$$\sqrt{n}(\hat{\beta} - \beta_0) = O_p(1), \quad \sqrt{n}(\hat{\alpha}_0 - \alpha_{0,0}) = O_p(1),$$

and the first-order Taylor expansion of the estimation error gives:

$$d_{\alpha_j}(\hat{\theta}) - d_{\alpha_j}(\theta_0) = \frac{\partial^2 l}{\partial \alpha_j\, \partial \beta}(\hat{\beta} - \beta_0) + \frac{\partial^2 l}{\partial \alpha_j\, \partial \alpha_0}(\hat{\alpha}_0 - \alpha_{0,0}) + o_p(\sqrt{n}).$$

By the block-diagonal structure, the expected values of both cross-second-derivative terms are zero. After normalising by $\sqrt{n}$, the estimation error is $o_p(1)$. The centered term $d_{\alpha_j}(\theta_0)/\sqrt{n}$ dominates, CLT applies, and the χ^2 limit goes through.

This is why BP do not simply cite the "general score result": the general theorem assumes a fully specified measure. BP need to carry out the estimation-error accounting explicitly.

The Probabilist's three steps in the BP proof

The Probabilist identifies the BP proof as three distinct steps.

Step 1: Simplify the general LM statistic for block-diagonal $\mathcal{J}$. The full LM is $d'\mathcal{J}^{-1}d$. Block-diagonal structure reduces this to

$$\mathrm{LM} = \hat{d}_\alpha' \hat{\mathcal{J}}_{\alpha\alpha}^{-1} \hat{d}_\alpha,$$

where $\hat{d}_\alpha$ is the score in the α-directions only, evaluated at $\hat{\theta}$. The nuisance block (β, α_0) drops out because $\hat{d}_\beta = \hat{d}_{\alpha_0} = 0$ at the constrained MLE by the first-order conditions.

Step 2: Evaluate the simplified statistic. Substituting the BP model and computing $\hat{d}_\alpha$ and $\hat{\mathcal{J}}_{\alpha\alpha}$ explicitly (with h' terms cancelling), the LM statistic equals $\frac{1}{2}\widehat{\mathrm{ESS}}$ from the auxiliary regression of $g_t = \hat{u}_t^2/\hat{\sigma}^2$ on z_t. This is BP's Theorem.

Step 3: Apply Amemiya's CLT for regression coefficients. Let $\hat{\delta}$ be the OLS coefficients in the auxiliary regression of g_t on z_t, and $S = \hat{\delta}'(Z'Z)\hat{\delta}$ the explained sum of squares. Amemiya (1977) shows that under H_0:

$$\sqrt{n}(\hat{\delta} - 0) \;\xrightarrow{d}\; \mathcal{N}(0,\; 2\sigma^4 D^{-1}), \qquad D = \lim n^{-1} Z'Z.$$

The coefficient $\hat{\delta}$ is asymptotically centred at zero because g_t is asymptotically uncorrelated with z_t under H_0 — this is the centration argument resolved: it is not the score itself that is shown to be centered, but the auxiliary regression coefficient. From this CLT it follows that $(2\hat{\sigma}^4)^{-1}S \xrightarrow{d} \chi^2_{p-1}$, which equals χ^2_q in our notation.

Remark 23 (Why asymptotic centration works). Boris conjectured that "maybe there will be some asymptotic centration." This is exactly right, but the route is indirect. Under H_0, $g_t = \hat{u}_t^2/\hat{\sigma}^2 \approx \varepsilon_t^2/\sigma^2$, which has mean 1 and is independent of z_t (since ε_t are i.i.d. and z_t are fixed design points). The auxiliary regression of g_t on z_t therefore has true slope zero — this is the centration. It is asymptotic because it uses $\hat{u}_t$ rather than u_t, and the approximation $\hat{u}_t \approx u_t$ is valid only for large n.

Remark 24 (Why BP do not invoke the general score theorem). The general theorem states: if $T_n = n^{-1/2} \sum_t \partial \log p_t/\partial\gamma \xrightarrow{d} \mathcal{N}(0, \mathcal{J})$ under H_0 (with γ the tested parameter), then $\mathcal{J}^{-1/2}T_n$ has components that are asymptotically i.i.d. $\mathcal{N}(0, 1)$, and $\|\mathcal{J}^{-1/2}T_n\|^2 \xrightarrow{d} \chi_q^2$. The theorem requires that $\mathcal{J}$ and T_n be evaluated at the *true* parameter. With unknown nuisance, one must show that substituting $\hat{\theta}$ for θ_0 does not change the limit — and this is precisely what Steps 1–3 above establish. The general theorem provides the framework; BP's three steps provide the model-specific verification.